

Volumetric JEPA with VICReg for Cross-Modal Brain MRI Retrieval

A multi-view subject-level strategy for the ceT1-to-T2 retrieval challenge

Team proposal note

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Abstract

The challenge is to retrieve, for each query brain MRI volume in post-contrast T1 (ceT1), the matching T2 target volume from a same-dataset gallery. Only **dataset1** provides labelled ceT1-T2 training pairs. **dataset2** and **dataset3** provide no labelled training pairs and test generalization to, respectively, independent geometric deformations and preoperative-to-intraoperative structural changes.

We propose to convert **dataset1** into a multi-view subject-level training set. For each labelled subject, we generate original, geometry-deformed, non-linearly deformed, and pseudo-intraoperative target views. We then train a volumetric VideoJEPA-like model that predicts target T2 latent slice tokens from query ceT1 latent slice tokens. VICReg regularization is added to prevent collapse and to structure the latent space. At inference, each query-target candidate pair is scored by cumulative latent JEPA prediction error across slices or 3D layers. Targets are ranked by increasing latent energy.

1 Challenge objective

For each query volume q in ceT1 modality, the system must rank all T2 candidate volumes g from the corresponding same-dataset and same-split gallery:

$$q \mapsto (g_{(1)}, g_{(2)}, \dots, g_{(N)}),$$

where the true same-subject T2 target should appear as high as possible. The evaluation metric is mean reciprocal rank (MRR):

$$\text{RR}(q) = \frac{1}{\text{rank}_q(g^*)}, \quad \text{MRR} = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \text{RR}(q).$$

The final score is the average of the MRRs computed on the three independent datasets:

$$\text{Score} = \frac{\text{MRR}_1 + \text{MRR}_2 + \text{MRR}_3}{3}.$$

Key interpretation

The task is not segmentation or classification. It is a cross-modal retrieval problem: learn a representation or energy function that recognizes the same subject across ceT1 and T2, even when geometry or anatomy changes.

2 Data regime and generalization problem

The most important constraint is that we do *not* know the hidden matches in **dataset2** or **dataset3**. Therefore, the training signal must come from **dataset1**. The goal is to make the model robust enough to generalize to the two target regimes.

Dataset	Labelled training pairs	Main difficulty	Target behavior
dataset1	Yes, 350 pairs	Registered preoperative ceT1-T2	Same grid, easier matching
dataset2	No	Rigid and non-linear deformations	Query and target deformed independently
dataset3	No	Preoperative-to-intraoperative shift	Tissue shift, missing parts, local omissions

Table 1: The labelled source is **dataset1**. The other datasets evaluate generalization.

3 Main idea: make **dataset1** a multi-view subject-level dataset

For each labelled pair in **dataset1**, we know that

$$(q_i, t_i) = (\text{ceT1}_i^{\text{pre}}, \text{T2}_i^{\text{pre}})$$

belongs to the same subject i . We use this subject identity as the only reliable supervision.

We generate multiple views of the same subject:

$$\mathcal{V}_i^q = \{q_i^{\text{orig}}, q_i^{\text{geo}}, q_i^{\text{def}}, q_i^{\text{int}}\},$$

$$\mathcal{V}_i^t = \{t_i^{\text{orig}}, t_i^{\text{geo}}, t_i^{\text{def}}, t_i^{\text{pseudo-intra}}\}.$$

All views keep the same subject ID i . Therefore, any pair

$$(q_i^a, t_i^b), \quad q_i^a \in \mathcal{V}_i^q, \quad t_i^b \in \mathcal{V}_i^t$$

is a positive pair, while pairs with different IDs are negatives.

3.1 Dataset2-like augmentation

To mimic **dataset2**, we apply independent 3D geometric transformations to the query and target:

$$\tilde{q}_i = A_q(q_i), \quad \tilde{t}_i = A_t(t_i),$$

where A_q and A_t are sampled independently. Possible transformations include:

- 3D rotations and translations;
- small scaling and anisotropic zoom;
- B-spline or elastic non-linear deformation;
- random crop, padding, and resampling;
- moderate intensity perturbations.

The independence of A_q and A_t is important: it imitates the setting where the correct ceT1 and T2 no longer share a common geometry.

3.2 Dataset3-like augmentation

To mimic **dataset3**, we generate pseudo-intraoperative or postoperative-like T2 targets:

$$\hat{t}_i^{\text{intra}} = G(t_i^{\text{pre}}).$$

The transformation G should preserve subject identity while perturbing local structures. It may be implemented as:

- a surgical simulation augmentation: local cutout, cavity simulation, local displacement, elastic deformation;

- a lightweight VAE or autoencoder trained on unlabelled intraoperative T2 targets, if allowed by the competition rules;
- a generator that projects preoperative T2 volumes toward an intraoperative-like distribution.

This view is a synthetic positive, not an exact ground truth. Therefore, it should receive a smaller training weight than original or geometric positives.

No hidden labels are used

The hidden matches in **dataset2** and **dataset3** are not known. They should not be used for training. The proposed generalization strategy relies on labelled **dataset1** pairs and synthetic transformations whose subject identity is known by construction.

4 Volumetric JEPA model

A 3D MRI volume can be treated as an ordered sequence of slices or 3D blocks:

$$x = (x_1, x_2, \dots, x_K).$$

This is analogous to a video, where the slice axis replaces the temporal axis. The encoder maps the volume to latent tokens:

$$E_\theta(x) = (z_1, z_2, \dots, z_K), \quad z_k \in \mathbb{R}^d.$$

A global subject embedding is obtained by pooling:

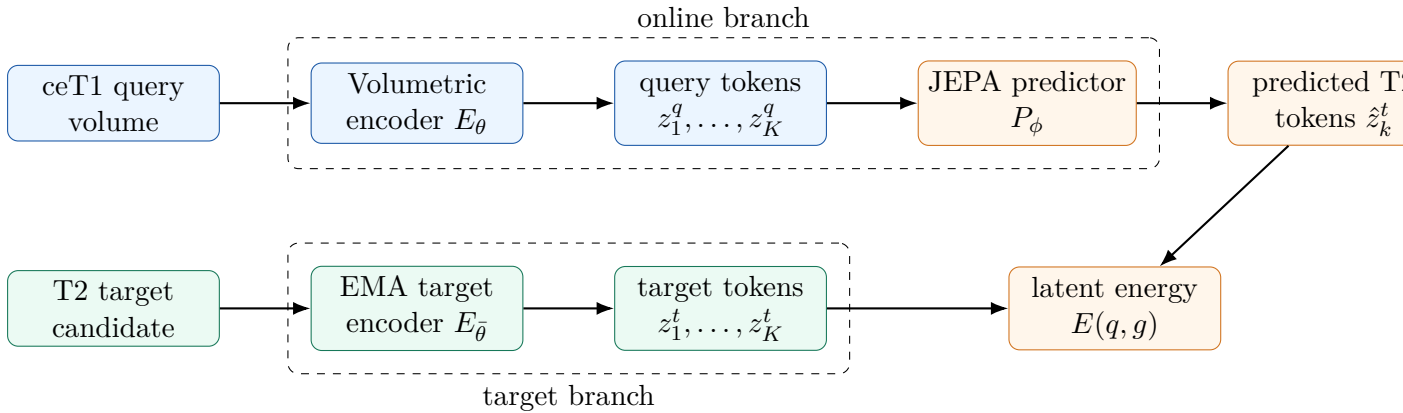
$$h(x) = \text{Pool}(E_\theta(x)).$$

The target branch uses an EMA or stop-gradient encoder:

$$E_{\bar{\theta}}(x) = \text{sg}(E_\theta(x)),$$

where $\bar{\theta}$ is updated as an exponential moving average of θ .

4.1 Architecture sketch



5 Training losses

5.1 JEPA latent prediction loss

For a positive pair (q_i^a, t_i^b) , the online encoder and predictor produce predicted T2 tokens:

$$\hat{z}_{i,k}^t = P_\phi(E_\theta(q_i^a)_k).$$

The target tokens are encoded with stop-gradient:

$$z_{i,k}^t = \text{sg}\left(E_{\hat{\theta}}(t_i^b)_k\right).$$

For aligned pairs, the basic JEPA loss is

$$\mathcal{L}_{\text{JEPA}} = \frac{1}{K} \sum_{k=1}^K \left\| \hat{z}_{i,k}^t - z_{i,k}^t \right\|_2^2.$$

For deformed or intraoperative-like views, exact slice alignment may fail. A robust version uses local matching:

$$\mathcal{L}_{\text{JEPA}}^{\text{win}} = \frac{1}{K} \sum_{k=1}^K \min_{\ell \in \mathcal{W}(k)} \left\| \hat{z}_{i,k}^t - z_{i,\ell}^t \right\|_2^2,$$

where $\mathcal{W}(k)$ is a small window of neighboring slices. A differentiable soft-min can replace the hard minimum:

$$\text{softmax}_{\ell \in \mathcal{W}(k)} d_{k\ell} = -\tau_s \log \sum_{\ell \in \mathcal{W}(k)} \exp(-d_{k\ell}/\tau_s).$$

5.2 Multi-positive retrieval loss

In a batch, each query view from subject i has several positive target views from the same subject:

$$\mathcal{P}(q_i) = \left\{ t_i^{\text{orig}}, t_i^{\text{geo}}, t_i^{\text{def}}, t_i^{\text{pseudo-intra}} \right\}.$$

All target views from other subjects are negatives. Let h_q and h_t be global embeddings. We define

$$S(q, t) = \frac{\cos(h_q, h_t)}{\tau}.$$

The multi-positive retrieval loss is

$$\mathcal{L}_{\text{ret}}(q_i) = -\log \frac{\sum_{t \in \mathcal{P}(q_i)} w_t \exp(S(q_i, t))}{\sum_{t \in \mathcal{B}_t} \exp(S(q_i, t))},$$

where $\mathcal{B}_t$ is the set of target views in the batch and w_t can downweight synthetic pseudo-intraoperative targets.

This is preferable to a fixed patient-ID classifier. A classifier would learn the 350 training identities, whereas retrieval training learns the relative rule: same subject pairs should be close, different subject pairs should be far.

5.3 VICReg regularization

VICReg is added to avoid collapse and structure the latent space. For two positive global embeddings h_q and h_t , the invariance term is

$$\mathcal{L}_{\text{inv}} = \|h_q - h_t\|_2^2.$$

The variance term prevents dimensions from becoming inactive:

$$\mathcal{L}_{\text{var}}(H) = \frac{1}{d} \sum_{j=1}^d \max(0, \gamma - \sigma(H_{:,j})),$$

where H is a batch of embeddings and $\sigma(H_{:,j})$ is the batch standard deviation of dimension j . The covariance term decorrelates latent dimensions:

$$\mathcal{L}_{\text{cov}}(H) = \frac{1}{d} \sum_{a \neq b} \text{Cov}(H)_{ab}^2.$$

Thus,

$$\mathcal{L}_{\text{VICReg}} = \lambda_I \mathcal{L}_{\text{inv}} + \lambda_V \mathcal{L}_{\text{var}} + \lambda_C \mathcal{L}_{\text{cov}}.$$

5.4 Final objective

The final training objective is

$$\mathcal{L} = \lambda_{\text{jepa}} \mathcal{L}_{\text{JEPA}} + \lambda_{\text{ret}} \mathcal{L}_{\text{ret}} + \lambda_{\text{vic}} \mathcal{L}_{\text{VICReg}}$$

with separate weights for original, geometry-deformed, and pseudo-intraoperative positives:

$$\mathcal{L}_{\text{jepa}} = \mathcal{L}_{\text{orig}} + \alpha_{\text{geo}} \mathcal{L}_{\text{geo}} + \alpha_{\text{pseudo}} \mathcal{L}_{\text{pseudo}}, \quad \alpha_{\text{pseudo}} < \alpha_{\text{geo}} \leq 1.$$

6 Inference: retrieval by cumulative latent JEPA error

At inference, we precompute target tokens for every gallery T2 volume:

$$Z_g = E_{\bar{\theta}}(g) = (z_1^g, \dots, z_K^g).$$

For each query ceT1 volume, the model predicts target T2 tokens:

$$\hat{Z}_q = P_{\phi}(E_{\theta}(q)) = (\hat{z}_1^q, \dots, \hat{z}_K^q).$$

The simplest latent energy is

$$E(q, g) = \sum_{k=1}^K \|\hat{z}_k^q - z_k^g\|_2^2.$$

For deformed or intraoperative cases, a windowed score is more robust:

$$E_{\text{win}}(q, g) = \sum_{k=1}^K \min_{\ell \in \mathcal{W}(k)} \|\hat{z}_k^q - z_{\ell}^g\|_2^2.$$

For **dataset3**, where some local structures may be missing or strongly changed, a trimmed energy can be used:

$$E_{\text{trim}}(q, g) = \sum_{k \in \text{Best}_{p\%}} e_k(q, g),$$

where e_k are per-slice errors and only the best $p\%$ of slices are kept. This prevents a local surgical cavity from dominating the score.

A combined score can mix latent prediction error and global embedding similarity:

$$\text{Score}(q, g) = E_{\text{win/trim}}(q, g) + \beta (1 - \cos(h(q), h(g))).$$

Candidates are ranked by increasing score.

7 Algorithm

Training algorithm

1. Load labelled pairs from `dataset1/train_pairs.csv`.
2. For each pair (q_i, t_i) , generate same-subject views:
 - original pair (q_i, t_i) ;
 - independent geometry-deformed pair $(A_q(q_i), A_t(t_i))$ for `dataset2`-like training;
 - pseudo-intraoperative target $(q_i, G(t_i))$ for `dataset3`-like training.
3. Encode query views with the online volumetric encoder E_θ .
4. Encode target views with the EMA target encoder $E_{\hat{\theta}}$.
5. Train the predictor P_ϕ with latent JEPA prediction loss.
6. Add multi-positive retrieval loss across subject IDs in the batch.
7. Add VICReg to prevent collapse and decorrelate latent dimensions.
8. Update the target encoder by EMA.

Inference and submission algorithm

1. For each dataset split, load the corresponding query CSV and gallery CSV.
2. Precompute T2 gallery tokens and global embeddings.
3. For each ceT1 query, predict T2 latent tokens with the JEPA predictor.
4. For every candidate target in the same gallery, compute cumulative latent prediction error.
5. Optionally use windowed or trimmed scoring for `dataset2` and `dataset3`.
6. Sort targets by increasing energy.
7. Write one row per query: `query_id,target_id_ranking`.

8 Expected strengths and risks

8.1 Strengths

- The model uses the volumetric nature of MRI: slices are treated as ordered layers, as in a video.
- The method is aligned with the challenge metric: retrieval is performed by ranking candidate targets.
- Dataset2 robustness is trained explicitly through independent geometric transformations.
- Dataset3 robustness is encouraged through pseudo-intraoperative target generation and trimmed latent scoring.
- VICReg reduces the risk of representation collapse, which is important for JEPA-style training.

8.2 Risks

- If pseudo-intraoperative generation is too aggressive, subject identity may be destroyed.
- Too much invariance can remove patient-specific anatomical cues needed for retrieval.
- Exact slice-wise scoring may fail on deformed volumes, hence windowed or attention-based matching is recommended.
- Training a large 3D model may be expensive. A slice encoder plus axial transformer is a practical compromise.

9 Minimal practical version

A hackathon-friendly implementation can be simplified as follows:

1. Resample or crop volumes to a manageable common size for batching.
2. Use a 2D CNN or small ViT slice encoder shared across slices.
3. Add axial positional embeddings and a small Transformer over slice tokens.
4. Train on `dataset1` with original and geometry-deformed pairs.
5. Add JEPA latent prediction plus batch-wise retrieval loss.
6. Add VICReg on global embeddings.
7. Rank each query by cumulative latent JEPA prediction error.

One-sentence summary

We turn the only labelled set into a multi-view subject-level training set, train a volumetric JEPA with VICReg to predict target T2 latent layers from query ceT1 layers, and perform retrieval by ranking gallery candidates according to cumulative latent prediction error.